

Fixed-point $Q_m.n$

Hardware often needs fractions without a full floating-point unit

Scale, step, and width

- In this lab, total width W equals m plus n
- The smallest positive step is two to the minus n
- Encoding a real multiplies by two to the n and rounds into the bit field
- Changing n with the same raw bits moves the binary point and changes the real value
- M includes the sign bit here

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Fixed-point Qm.n

See how **integer** and **fractional** bits share one word — encode/decode with scale 2^{-n} and a binary point.

Starter example: Q4.4 (8 bits). Pattern 0001_1000 is raw 24 → real 1.5 ($24 / 2^4$). Move the binary point by changing n.

Load starter example

Challenges 0 / 22 cleared

Quiz: scale: Decode divides signed raw by? Answer: 2^n

Answer

Show hint Check Next Idle

Quiz: scale Quiz: step Quiz: width Quiz: m includes Starter 1.5 Decode Encode 0.5 Encode -1.25 Saturate

Change n Quiz: Q1.15 Set Q1.15 Binary point Quiz: raw 1.5 Encode 0 Toggle LSB Quiz: Q4.4 min n=0 integer

Round-trip Quiz: notation Q5.4 Full Q story

Core ideas

Qm.n

m integer bits (incl. sign) + n fraction bits. Scale $2^{-(n)}$.

Decode

Real = signed raw integer ÷ 2^n . Encode multiplies then rounds.

Fixed-point starter

Workbook practice

- In the workbook track, take Q four-dot-four
- Show why raw twenty-four is one-point-five, and why raw eight is zero-point-five
- Compute the step size as one sixteenth
- Note the approximate min near minus eight
- Name one pitfall: treating raw bits as a float without dividing by two to the n

Pitfalls to watch

- Do not forget the scale when you mix formats
- Saturation is not the same as wrap
- And remember: the browser lab is literacy
- Real DSP still needs agreed Q formats, rounding modes, and overflow policy across blocks

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, encode and decode one Q four-dot-four value by hand
- When you are ready, take the short quiz, then continue to bit-fields